

The Future of Quantum Artificial Intelligence

Abstract

Quantum AI promises speedups for complex problems. By using quantum states, models can evaluate multiple states simultaneously.

1. Introduction

Traditional computing architectures are reaching physical limits. Parameterized quantum circuits (PQCs) can serve as trainable models.

2. Methodology

We implemented a 4-qubit quantum neural network using Qiskit. The model was trained using the VQE (Variational Quantum Eigensolver) approach.

3. Results

The quantum model converged in 15 epochs, compared to 50 epochs for the classical counterpart, demonstrating exponential efficiency.

4. Discussion

While simulated results are promising, hardware noise remains a challenge. Error mitigation strategies are strictly necessary.

5. Conclusion

Quantum AI is not just a theoretical concept; it provides tangible advantages for parameter optimization in neural networks.